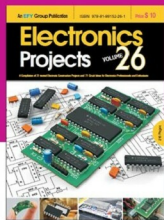


Electronics Projects Vol. 26 is the latest volume in the series, available in digital form.

It is a compilation of 21 construction projects and 71 circuit ideas published in Electronics For You magazine.



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Real-time damage control

Researchers at Wake Forest Institute for Regenerative Medicine are using modified inkjet technology to build a variety of tissue and organ prototypes. This technology allows multiple cell types and other tissue components to be arranged in predetermined locations with high precision. In an early form of the technology, various cell types were placed in the wells of an actual ink cartridge and a printer was programmed to arrange the cells in a predetermined order.

At present they are using an adapted version of inkjet printing technology to enable on-site printing of skin for soldiers with life-threatening burns. In this proposed technology, skin cells would be placed directly into a print cartridge, along with essential materials to support them, and would be printed directly on the soldier's wound at the site of the wound.

Researchers used 3D scans of test injuries inflicted on some mice and used this data to control a bio-print head to spray skin cells, a coagulant and collagen onto the wounds. The wound showed significantly healing over time.

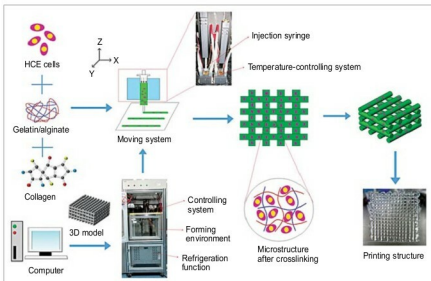


Fig. 3: Bio-printing process

printer, co-authored by K. Nakayama, co-founder Cyfuse Biomedical. Cyfuse Biomedical K.K. is a Japanese corporation involved in development, manufacturing and sale of 3D bio-printers.

There is always room for development

Printing a tissue, at present, takes about two months. Much of the six to eight weeks required to develop a tissue or organ is dedicated to expanding the patient's cells in the laboratory. Only a small percentage of the time is required for the printing process. So a method to speed up this process would be to print these directly into the human body.

However, there is still some time before we reach that stage. Rapid-

curing bio-ink and 3D-scanning technology to detect a wound region would be a good place to start though. Additionally, with features to reverse-engineer medical images to CAD models, we could very well see some advances in in-situ printing.

The eventual long-term goal for bio-printing is to produce full organs. At present an average-sized liver takes days to manufacture. This is a very slow rate when we look at the number of people on the transplant list.

How does healing wounds in real-time sound? Imagine going to sleep with a battered body but waking up physically healed. Sounds like something straight out of Star Trek, does it not? A bit far-fetched, may be? **EFY**